

1. Scenario: Word Cloud for Popular Keywords

Question: You're given a collection of product reviews. How will you create a word cloud to visualize the most frequent keywords?

- Combine all the reviews into one large text.
- Remove stopwords (like the, and, is) and punctuation.
- Convert all text to lowercase.
- Split the text into words and count how often each word appears.
- Create a word cloud where bigger words = more frequent words.

2. Scenario: Sentiment Analysis for Hotel Reviews

Question: A hotel wants to understand how customers feel about their service. How would you apply sentiment analysis to the reviews?

- Collect all hotel reviews in a dataset.
- Clean the text (remove stopwords, punctuation, special symbols).
- Convert text into numbers using a vectorizer (CountVectorizer or TF-IDF).
- Use a sentiment model or lexicon (like VADER or TextBlob).
- Label each review as Positive, Negative, or Neutral.
- Calculate the overall sentiment percentage for the hotel.

3. Scenario: Topic Modeling for News Articles

Question: You're given thousands of news articles with no labels. How would you identify the main themes or topics?

- Collect all news articles into one dataset.
- Clean the text (remove stopwords, punctuation, lowercase).
- Convert text into numerical form using TF-IDF or CountVectorizer.
- Apply LDA (Latent Dirichlet Allocation) or NMF.
- Each topic will show a group of important words.
- Name each topic based on those keywords.

4. Scenario: Feature Extraction Using CountVectorizer

Question: You want to feed customer messages into a machine learning model. How do you convert the text into numerical features using CountVectorizer?

- Collect all customer messages.
- Clean the text (lowercase, remove punctuation and stopwords).
- Use CountVectorizer() to build a vocabulary.
- Transform the text into a matrix of word counts.
- Each row = one message, each column = one word.
- Feed this matrix into your machine learning model.

5. Scenario: Choosing Between CountVectorizer and TF-IDF

Question: When analyzing job descriptions, should you use CountVectorizer or TF-IDF Vectorizer? Why?

- Use CountVectorizer when:
You only care about how often words appear.

- Use TF-IDF when:
You want important and unique words.
You want to reduce the impact of common words like job, work, company.

So, TF-IDF is better for job descriptions because it highlights meaningful skills and removes noise.

6. Scenario: Using TF-IDF to Find Important Terms in a Resume

Question: You want to extract the most unique and relevant words from a resume. How would TF-IDF help?

How TF-IDF helps:

- Clean the resume text (remove stopwords, lowercase, punctuation).
- Apply TF-IDF Vectorizer.
- Words that:
Appear often in the resume, But rarely in other resumes → get higher scores.
- Sort the words by TF-IDF score.
- The top words are the most unique and important skills.